User Manual – Solar Gong Valve Controller

Introduction

Hello friend, thank you very much for your order. I hope you will find it useful and according to your expectations. There are some instructions before you start.

Device Handling and Safety

This device is fully functional, and you must be **very careful when opening the cover** to look inside. If you break something, it may be **impossible or very difficult to repair**, because the components are tightly fixed together. **Handle it gently.**

Connecting to the Serial Monitor

To understand the device, you need to open it and **connect a micro-USB cable**.
Download the Arduino IDE, and from there you can **check the serial monitor**. You can view the serial data to see how the device is working. The monitor will show:

- The **current mode**
- The **light value (lux)**
- The **threshold level**
- The **flag status**
- All **device logs**

This helps you understand how the device operates in real time.

Battery and Solar System

This device works on solar power. The battery inside is a **$LiFePO_4$ battery**, which gives good backup. It is continuously charged through the **MPPT solar converter module**, which converts solar energy to electrical energy and charges the battery.

The module has **four LEDs** that indicate the battery level from **25% to 100%**. You don't need to check it often after installation, but it is available if needed.

There is also a **setting mode on the solar module** to select the correct battery type. It is already set to **$LiFePO_4$**.

Inside the device, you will see a **female socket** near the battery for **manual charging** if the voltage becomes low. I am sending the **adopter and socket** so you can charge it manually whenever needed.

Front Indicators and Buttons

On the front panel, there is an **LED indicator**.
When you turn on the device from the bottom switch, the **green light means the device is on**. When the device is not in use, **turn off the main power button**.

Next to the power button at the bottom, there is a **small push button**. This is **not** the power button. This button is used to **select different modes** of the device.

Solenoid Valve Manual Test

When you turn on the device and **press the mode button for the first time**, it will **turn on the solenoid valve**.
Press it again to **turn the valve off**.

Be careful **not to keep the valve ON for too long**, because it uses around **3A** and will **drain the battery quickly**. Use this only for **short testing**, for example turn it on and after two seconds turn it off.

Calibration Mode

When you press the button, you will see the LED blink.
Press and release, and the **yellow LED will start blinking**, which means it is **looking for the input lux value**. This is the **calibration mode**.

When you press the button again, it **takes a snapshot of the current lux value**, stores that value, and updates the code. It saves the value in **flash memory**.

How the Algorithm Works

Here is how the logic works:

- The device has a **preset value of 50 lux** by default.
- When you calibrate, it **captures the current lux value** and stores it.
- The device keeps checking daylight.
- There is a **20-hour condition**.
- After 20 hours, the **flag resets automatically**.

Example:

- On the first day, the flag is **0**.
- You set the light value around a bright time, for example **11:30 am or 12 pm**.
- The sensor may read a lower value because it is **not fully exposed**, which is normal.
- If the sensor reads **200 lux**, it saves 200 lux.
- Later, at the brightest time, if the current lux becomes **greater than or equal** to the saved value:
    - It **triggers the solenoid valve for two seconds**
    - Turns it off
    - Sets the **flag to 1**

After 20 hours, the flag resets, and the cycle repeats.

Solar Connection and Calibration Check

You need to connect the **solar panel** and check the **calibration** to ensure proper operation.

Thanks and Best Regards,
Zaeem